

Ron's Retirement Finance Model V1

User Guide
Updated May 9, 2026

This guide describes the current single-file HTML model, including the newer survivor/widow(er) override, enhanced senior deduction handling, cash-flow funding lines, export behavior, and override-table workflow.

Data Retention: Read This First

The app stores your working data in your browser's local storage. That makes the app fast and private, but local browser storage is not a permanent backup. It can disappear if you clear site data, change browsers or devices, use a private/incognito window, reset browser settings, or return after a long time and the browser has purged local data.

Use two layers of backup

- **Set up GitHub sync if you want automatic backup.** The Export tab can connect the app to a private GitHub repository. When active, your data is saved to GitHub and cached locally in the browser. This is optional, but strongly recommended for long-term retention.
- **Export a JSON backup regularly.** Use Export tab -> Export JSON after meaningful changes, before testing alternate scenarios, and at least monthly. The JSON file is the portable copy you can import later if browser storage is lost.
- **Keep more than one JSON.** A date-stamped export lets you return to a known-good plan after experimenting with Roth conversions, survivor scenarios, tax table edits, or start-of-year actuals.
- **Import restores the saved model data.** Use Export tab -> Import JSON to restore a saved snapshot into the current browser. If GitHub sync was previously used, you may still need to reconnect the GitHub token because the token is stored only in this browser for security and is not included in exported JSON.

Suggested routine

- After initial setup: connect GitHub sync, then export a JSON backup.
- Before major scenario testing: export JSON, then make changes.
- After the scenario looks useful: export another JSON with the new assumptions.
- If the app opens blank or old after time away: import your latest JSON backup or reconnect/load from GitHub.

GitHub sync setup

The Export tab contains these same setup instructions and should be treated as the live in-app reference. GitHub occasionally changes labels or menu placement, but the basic process is: create a private repository, create a token with repo access, then connect it in the app.

Step	What to do
1. Create a GitHub account	Go to github.com and sign up for a free account if you do not already have one.
2. Create a private repository	Click the + icon in the top right, choose New repository, name it anything such as my-retirement-data, and set visibility to Private. Leave the other options at defaults and click Create repository. The app is designed for a private repo and rejects public repos.
3. Create a Personal Access Token	Click your profile photo, go to Settings, scroll to Developer settings, then choose Personal access tokens -> Tokens (classic) -> Generate new token (classic).

Step	What to do
Token details	Use a note/name such as retirement-model. For expiration, the app instructions suggest No expiration. Under scopes, check repo. Generate the token and copy it immediately; GitHub only shows it once.
4. Connect in the app	Return to the app's Export tab. Paste your GitHub username, repository name, and token, then click Connect & Verify.
After connection	Use Force Save Now to push immediately. The app stores plan data in finance_model_data.json in the private repo and keeps a local browser cache.

Security note: the GitHub token is stored only in the browser where you connected sync. It is not included in JSON exports. If you import your JSON backup on a new browser or device, reconnect GitHub sync there if you want automatic backup again.

Quick Start

- Use **Inputs** for plan-wide assumptions: names, birth years, filing status, state, growth, spending inflation, Social Security, Medicare/IRMAA, and employee contribution defaults.
- Use **Start of Year** for opening balances and year-specific actuals. Green fields are values you entered; blue fields are projected/default values.
- Use **Override Tables** to model a year range without changing the base plan. Leave blank fields blank; blanks mean no override.
- Use **Cash Flow**, **Tax Summary**, **Year Detail**, and **Dashboard Action Plan** to review what changed and why.
- Use **Export** to keep JSON and CSV backups. Export filenames include the plan names.

Tab	Purpose
Dashboard	High-level results, charts, and action-plan cards for the first projection years.
Inputs	Global assumptions and special-mode switches, including Widow(er) Override.
Start of Year	Opening balances and actual values that reset the projection from that year forward.
Override Tables	Targeted year-range changes for spending, taxes, conversions, growth, funding order, cash floor, HSA, and state tax.
Year Detail	Detailed view of one year: balances, income, taxes, funding, and actions.
Cash Flow	Full year-by-year table with sticky headers and frozen year/age columns.
Tax Summary	Focused tax view including filing status, deductions, senior bonus, income tax, IRMAA, and Medicare Part B.
Tax Tables	Editable federal, California, LTCG, IRMAA, Medicare Part B, standard deduction, and senior bonus parameters.
Export	GitHub sync, JSON backup/import, and projection CSV export.

Core Projection Flow

1. Mandatory income arrives first

Each year begins with start-of-year balances. The model then counts income that arrives regardless of discretionary funding needs: wages, self-employment income, Social Security, RMDs, inherited IRA draws, dividends/interest, and HSA withdrawals. Mandatory income funds spending first. Excess mandatory income is routed to securities.

2. Spending requirement is built

The spending requirement is Annual Spending plus Funded Tax plus any Cash Gap needed to rebuild cash to the cash floor. IRMAA and Medicare Part B are shown separately from income tax; they are Medicare costs rather than income tax funding.

3. Discretionary funding fills any remaining need

If mandatory income is not enough, the model draws cash down to the floor, then draws from securities or IRA depending on the funding-order setting, then uses Roth as a last resort. Funding order can be overridden by year range.

4. Taxes are solved iteratively

IRA draws, Roth conversions, taxable brokerage gains, dividends, Social Security taxation, and deductions interact with tax owed. The model repeats the funding/tax calculation until the tax funding draw and computed income tax converge to near-dollar precision.

5. End-of-year balances roll forward

Accounts grow, contributions and draws are applied, cash above the floor is swept to securities, and the resulting end-of-year balances become the next year's start-of-year balances unless overridden in Start of Year.

Dashboard Action Plan

- **Spending Requirement** now separates Annual Spending, Income Tax Funding, Cash Gap (floor rebuild), and Total Required.
- **Tax Picture** shows federal tax, state tax, total income tax, IRMAA surcharge, Medicare Part B, and effective/marginal rates.
- **Funding Surplus/Shortfall** highlights whether the year was over- or under-funded after the waterfall.
- **Why This Year Looks Different** flags unusual drivers such as tax override active, cash floor rebuild, Roth conversion, mandatory income surplus, funding shortfall, or survivor mode active.

Inputs Tab

Personal Information

- Names drive labels throughout the app and are used in export filenames.
- Filing status sets the default tax table view. Widow(er) Override can switch later years to Single without changing the base filing input.
- State currently supports California or Federal Only. A state tax override can model a year-range move to NONE or CA.
- Plan Start Year and Plan Length define the projection window.

Key Variables

Input	How it is used
Portfolio Growth Rate	Total return for invested accounts; can be overridden by year range.
Brokerage Dividend Rate	Dividend yield on securities. Brokerage dividends are taxable whether paid out or reinvested.
Cash Interest Rate	Interest on cash balances, taxable as ordinary income.
Reinvest Brokerage Dividends	If No, dividends are paid out as cash income. If Yes, they remain in brokerage and do not separately fund spending.
Dividend Tax Treatment	Ordinary or qualified/LTCG treatment for brokerage dividends.
Annual Spending Increase	Inflates spending after the anchor year unless spending is reset by SOY or override.
SS COLA	Inflates Social Security benefits after each person's start year.
Wage Growth	Default wage/self-employment growth; SOY entries and wage-growth overrides take priority.
Stock Sales Taxable Gain	Percent of securities sale proceeds treated as taxable capital gain.
Cash Floor	Minimum cash balance. If start cash is below floor, the model adds a Cash Gap to rebuild it.

Social Security

- Enter each person's start age and monthly benefit at that start age.
- SS End Year can still be used for manual scenarios. Use 9999 for indefinite benefits.
- When Widow(er) Override is active, the deceased spouse's SS End Year can be overridden automatically by the death year, and the survivor can receive the higher benefit.

Widow(er) Override / Survivor Modeling

This mode models one spouse dying in a chosen year and the survivor continuing the plan. It is designed for Roth-conversion and tax-rate analysis: the same household balances may later be taxed under Single brackets, with only one Medicare/IRMAA count and one Social Security benefit.

Setting	Behavior
Scenario	Off, Person 1 survives Person 2, or Person 2 survives Person 1.
Death Year	The year of death. The death year itself remains the original household setup, typically MFJ.
End deceased spouse Social Security	Starting the year after death, the deceased spouse's Social Security is set to zero.
Survivor receives higher Social Security benefit	Starting the year after death, the survivor receives the higher of the two modeled Social Security benefits, rather than simply keeping only their own benefit.
Switch survivor to Single filing	Starting the year after death, federal and CA brackets, standard deductions, taxable Social Security, Medicare/IRMAA count, and senior deductions use Single.

How balances are handled

- At the start of the first year after death, the deceased spouse's IRA balance is moved into the survivor's IRA bucket and the deceased spouse's Roth balance is moved into the survivor's Roth bucket.
- RMD calculations after death use the survivor's age/start-year rules for the inherited spouse balance now held by the survivor.
- Cash, securities, HSA, and inherited IRA remain household-level accounts and continue normally.
- The dashboard, cash flow, tax summary, year detail, and report hide the deceased spouse's age after the death year to reduce confusion.

Recommended workflow

- Set both SS End Years to 9999 for a normal lifetime baseline.
- Turn on Widow(er) Override, choose the survivor and death year, and keep the Social Security checkboxes on for the common survivor-benefit case.
- Compare Roth conversion years before and after the death year. This highlights whether converting more while MFJ reduces later Single-bracket/RMD pressure.
- Run alternate death years or survivor choices as separate JSON exports so you can compare scenarios.

Start of Year Tab

Start of Year entries are anchors. Entering a value for a year overrides the projection for that specific start-of-year field, and future years resume from the entered value.

- Use the first plan year to seed opening balances: cash, securities, IRAs, Roth accounts, HSA, and inherited IRA.
- Use later years to enter actual balances or income as time passes. This keeps the model aligned with reality without rewriting historical assumptions.
- Annual spending entered in Start of Year becomes the spending anchor for that year; later years inflate from it unless reset again.
- Gross wages, self-employment income, Social Security monthly amount, HSA deposits/withdrawals, and account balances can all be anchored by year.
- Blue fields represent projected/default values. Green fields represent user-entered values.

Override Tables

Override rows apply to a year range. Enter only fields you want to change. A blank amount is ignored; it does not mean zero. If ranges overlap, the last matching row wins. Use 9999 as End Year for forever.

Override	Use it for
Spending	Set a fixed spending amount, percentage change, or tax funding override for specific years.
Tax Deductions	Use itemized deductions when higher than standard deduction. Federal and CA can be entered separately. Non-itemizer charity requires an explicit Apply Charity opt-in and is capped by filing status.
IRMAA Level Target	Convert IRA to Roth to stay below a chosen Medicare MAGI threshold.
Income Level	Convert IRA to Roth up to a specific MAGI target. Useful for ACA/pre-Medicare or precise bracket planning.
Tax Bracket Fill	Convert IRA to Roth to fill the current federal bracket up to its ceiling minus headroom.
Growth Rate	Model down years, conservative periods, or different return assumptions.
Wage Growth	Model career-stage changes in wage or self-employment growth.
Funding Order	Choose whether securities or IRA is drawn first after cash and mandatory income.
Cash Floor	Raise/lower minimum cash balance by year range.
HSA Deposits	Schedule annual HSA contributions.
State Tax	Switch to CA, NONE, or supported future states by year range.

Roth Conversion Strategy

Priority when strategies overlap

Income Level takes priority over IRMAA targeting. IRMAA targeting takes priority over Tax Bracket Fill. The app warns about overlapping strategies so you can see which one is controlling the year.

IRMAA Level Target

- Level 0 means off. Higher levels correspond to Medicare MAGI thresholds in the tax tables.
- The model converts IRA to Roth up to the selected threshold minus headroom when possible.
- Fill mode can push toward the next tier. Roth mode can use Roth withdrawals to avoid exceeding the target when income is already too high.

Income Level

- Sets a dollar MAGI target. The model converts IRA to Roth up to but not above the target when IRA funds are available.
- This is the most precise way to model a one-year Roth conversion amount based on income.

Tax Bracket Fill

- The model finds the current federal bracket and converts IRA to Roth up to the top of that bracket minus headroom.
- Useful for filling low brackets before RMDs or before survivor-mode Single brackets begin.

Funding and Cash Flow Columns

Column	Meaning
Annual Spending	Modeled lifestyle spending for the year.
Funded Tax	Income tax funding required after withholding. A marker indicates a tax override.
Cash Gap	Amount needed to rebuild cash to the cash floor when the year starts below the floor.
Total Required	Annual Spending + Funded Tax + Cash Gap.
Total Funded	Mandatory income plus discretionary draws that fund the requirement.
Excess Income -> Securities	Mandatory income above the requirement, swept into brokerage.
Cash Sweep -> Securities	Cash above the floor moved to brokerage after settlement.
Shortfall	Amount the waterfall could not fund.

Tax Tables

Tax Tables are editable by year. If a future year is not configured, the app starts with the nearest prior table and applies the inflation helpers when you copy/add years.

- Federal ordinary income brackets and LTCG/qualified dividend brackets are editable by filing status.
- Federal standard deduction includes base deduction, senior add-on per person age 65+, the non-itemizer cash charity cap, and the enhanced senior bonus through 2028.
- Enhanced senior bonus is modeled per qualifying senior and phases out based on MAGI. Current table fields include Sr. Bonus, Phaseout Start, and Phaseout Max.
- California ordinary brackets and standard deduction are editable.
- IRMAA thresholds/surcharges and Medicare Part B base premium are editable and inflate by the table assumptions.
- Tax Summary shows filing status each year, which is useful when Survivor mode switches later years to Single.

Reports and Exports

- The Report button generates a printable projection report with assumptions, action plan, yearly projection rows, and active overrides.
- Manual JSON export saves the full app state. Use this before major scenario changes.
- JSON import replaces the current browser state with a saved snapshot.
- Projection CSV export is useful for spreadsheet analysis.
- GitHub sync, when connected, saves data to the configured private repository. Local browser storage remains the instant offline copy.

Assumptions and Limits

- This is a planning model, not tax, legal, investment, or Social Security advice.
- Federal, California, IRMAA, and Medicare rules can change. Review Tax Tables each year before relying on results.
- The model simplifies survivor/inheritance treatment to keep planning clear: retirement balances transfer into the survivor's matching IRA/Roth bucket at the start of the year after death.
- Brokerage sale tax uses a single taxable gain percentage. It does not track individual tax lots.
- IRMAA uses the two-year Medicare MAGI lookback and the tax table thresholds in the app.
- Tax overrides affect funding, while the tax picture still shows computed tax for analysis.
- The non-itemizer charity deduction is federal-only in this model. It applies only when the standard deduction is used, the override row has Apply Charity turned on, and only up to the tax-table cap for that filing status.